Zhongshan Institute of Advanced Cryogenic Technology

Experimental Validation of a Corrugated Membrane for Large-Scale Liquid Hydrogen Cargo Containment

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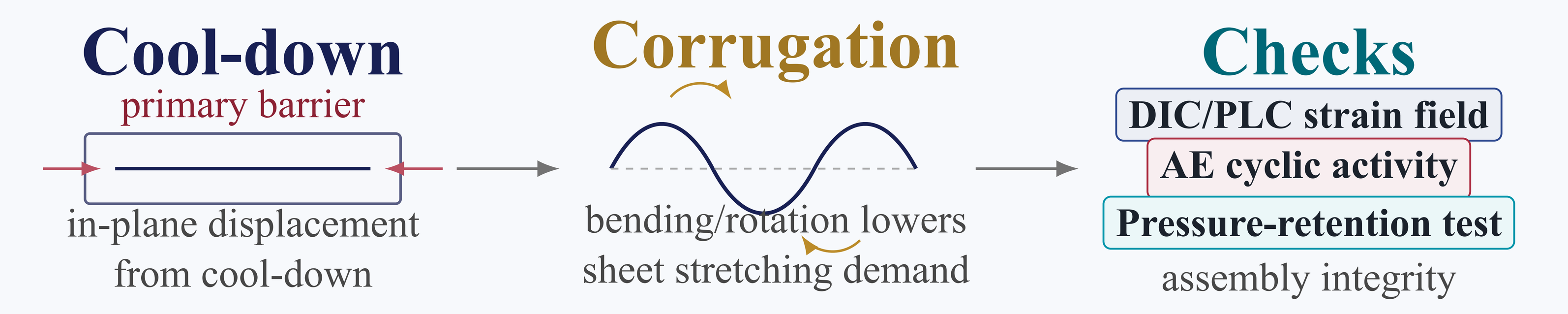
Introduction

Large-scale LH_2 cargo containment needs a metallic primary barrier that is leak-tight, formable, and tolerant of the in-plane displacement generated when the tank system cools from ambient temperature to cryogenic service. If this displacement is carried mainly by membrane stretching, local strain concentration can become a direct risk for cracking, leakage, and loss of containment reliability.

A corrugated 316L membrane is introduced to turn this constraint into a geometric compliance problem. The selected cross-corrugated profile is expected to accommodate imposed displacement through local bending and rotation of crests, valleys, and side-wall transitions, thereby lowering the uniform stretching demand on the thin sheet.

The remaining question is experimental rather than purely geometric: after patent-based forming, specimen preparation, and cyclic loading, does the membrane still follow the intended strain-release mechanism, and can an assembled module retain preliminary integrity after cryogenic cycling?

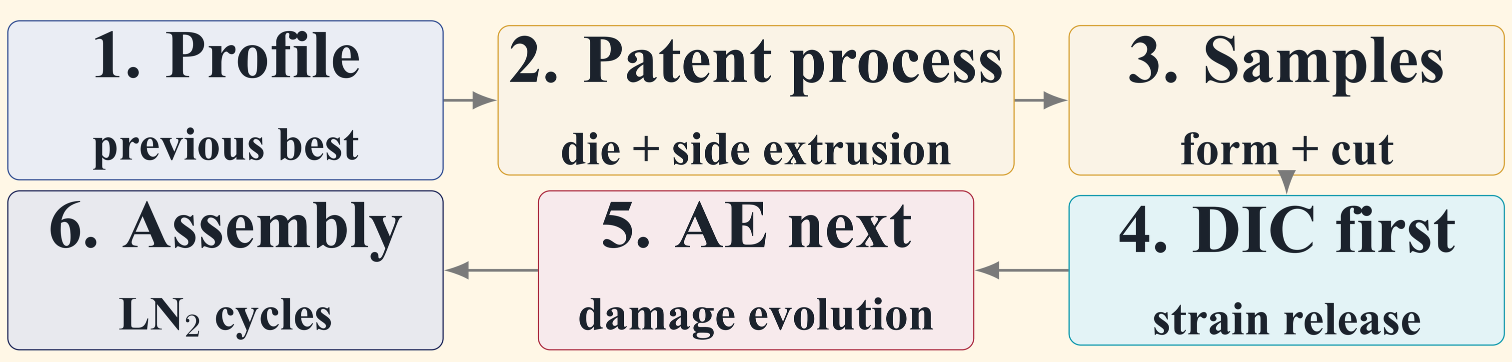
This poster therefore links three checks into one validation chain: DIC/PLC maps where imposed displacement is released, AE monitors cyclic damage activity during repeated loading, and pressure-hold plus helium leak testing are used as the pressure-retention method to verify assembly integrity after LN_2 fill/drain cycles.

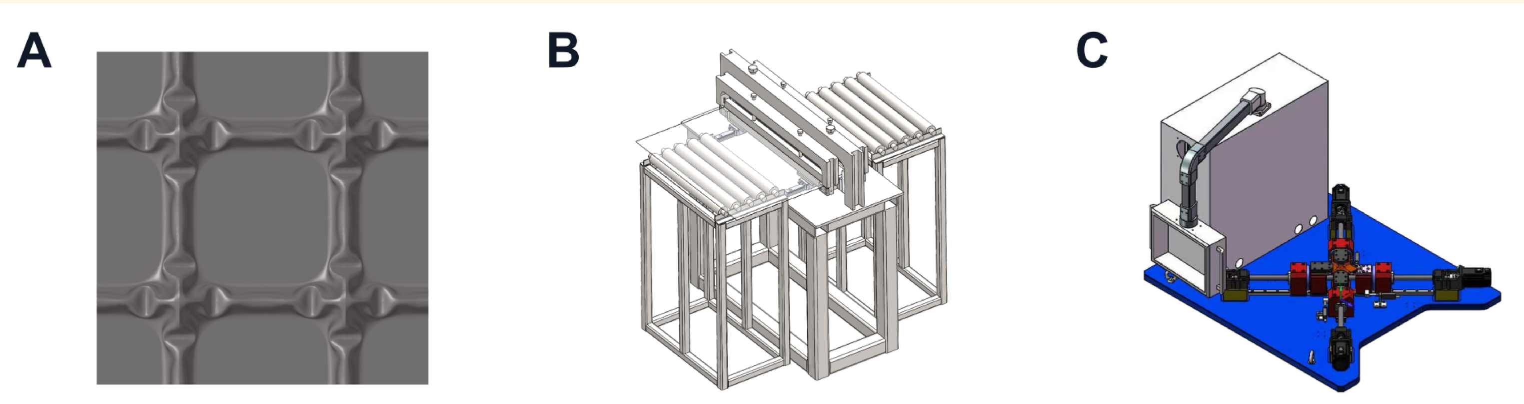


Conceptual schematic: cryogenic contraction sets the displacement demand; corrugation supplies the intended strain-release mechanism; pressure-retention testing checks assembled-membrane integrity after LN_2 cycling.

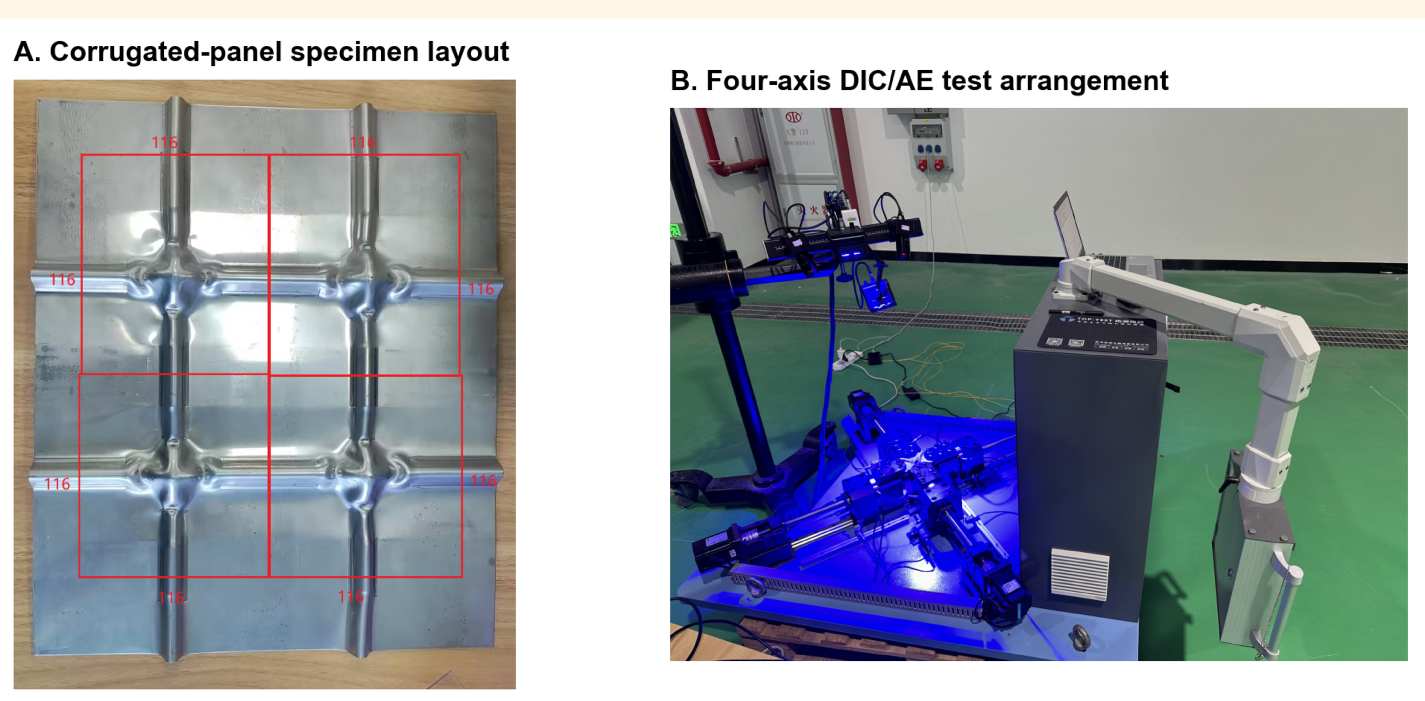
M. Methods: Design-to-Test Logic Chain

Starting point	Value	Why it matters
Selected profile	$e_r = 1.0$, $e_w = 5.0$, $\lambda = 0.651$	best-performing geometry from the previous design screen
Sheet and material	316L, $t = 1.2$ mm	thin metallic primary-barrier candidate
Patent basis	CN118237450A	coupled downward pressing and lateral squeezing for equal-section cross corrugation





Selected profile → patented processing machine → four-axis specimen platform. The process figure is used as design and preparation context, not as a dimensional CAD claim.



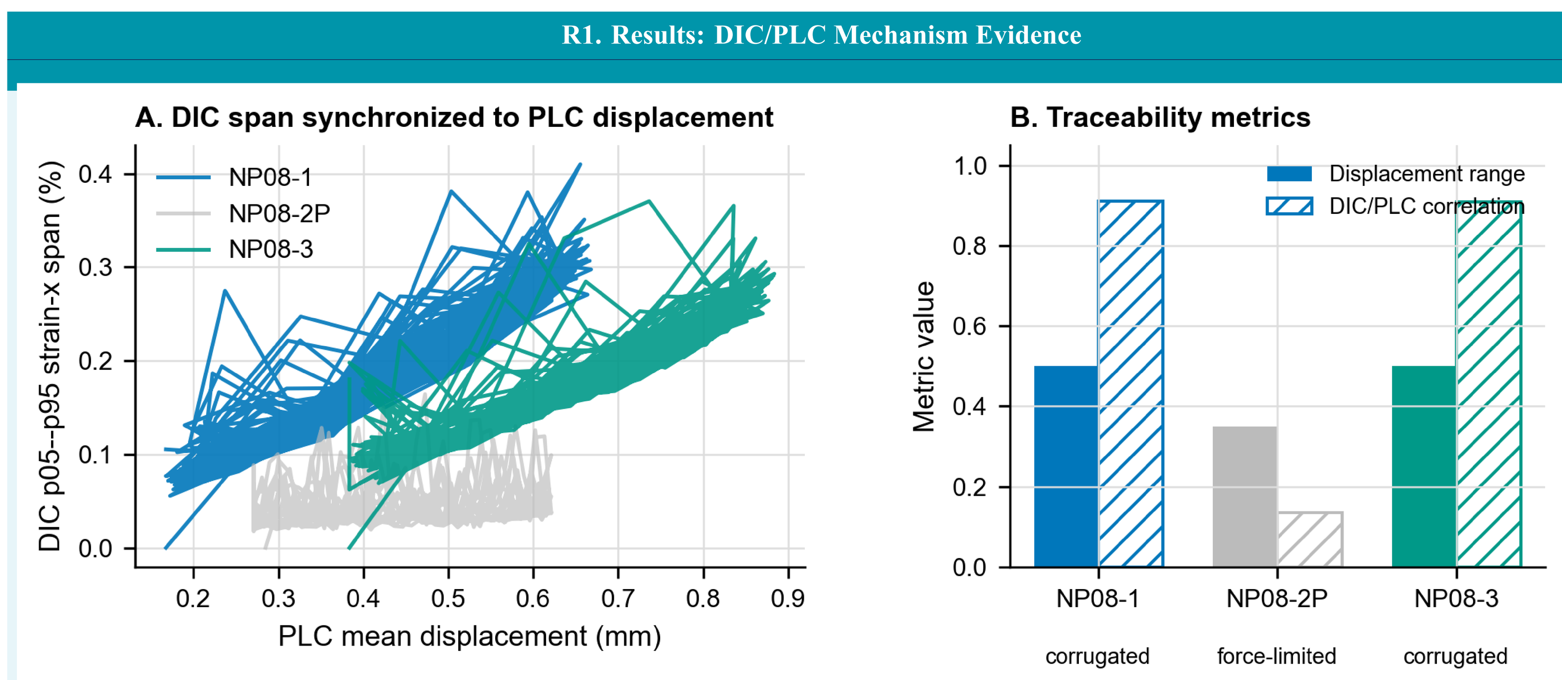
Specimen extraction, DIC field of view, and mechanical setup connect the formed profile to the DIC/AE tests.

Experiment	Direct question	Measured output
DIC/PLC	How does corrugation release imposed strain?	synchronized strain span, geometry-fixed ROI hierarchy
AE	When does cyclic activity concentrate?	relabeled cycle windows and damage-evolution screen
LN_2 assembly	Does pressure-retention testing verify assembly integrity after thermal cycling?	pressure-hold and helium leak checks across five LN_2 cycles

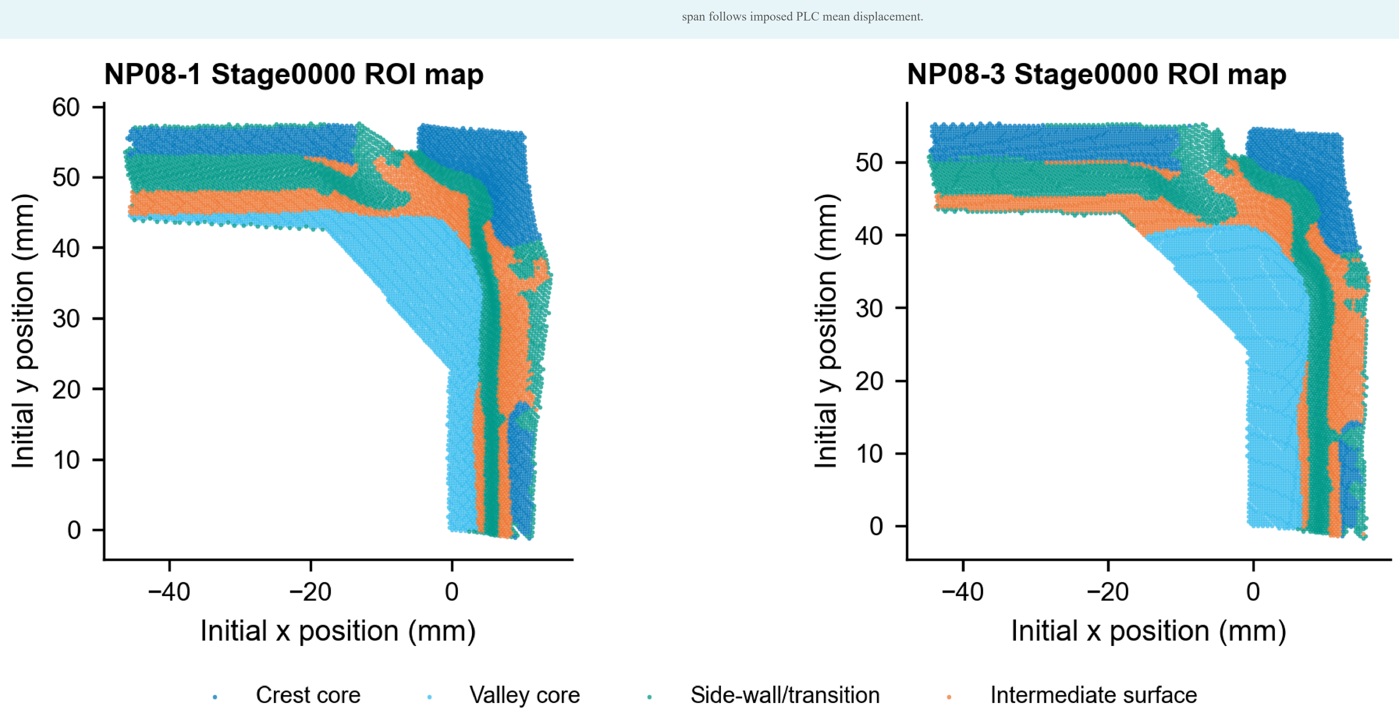
$$\bar{u}_k = \frac{1}{4}\sum_{a=1}^{4} u_{a,k}, \qquad S_i = \frac{dP_{95}(|\varepsilon_x|)_i}{d\bar{u}}, \qquad \Psi_i = A_i q_i^2$$

The evidence chain is deliberately ordered: profile and processing first, DIC mechanism second, AE damage-evolution screening third, and pressure-retention testing for assembly integrity last.

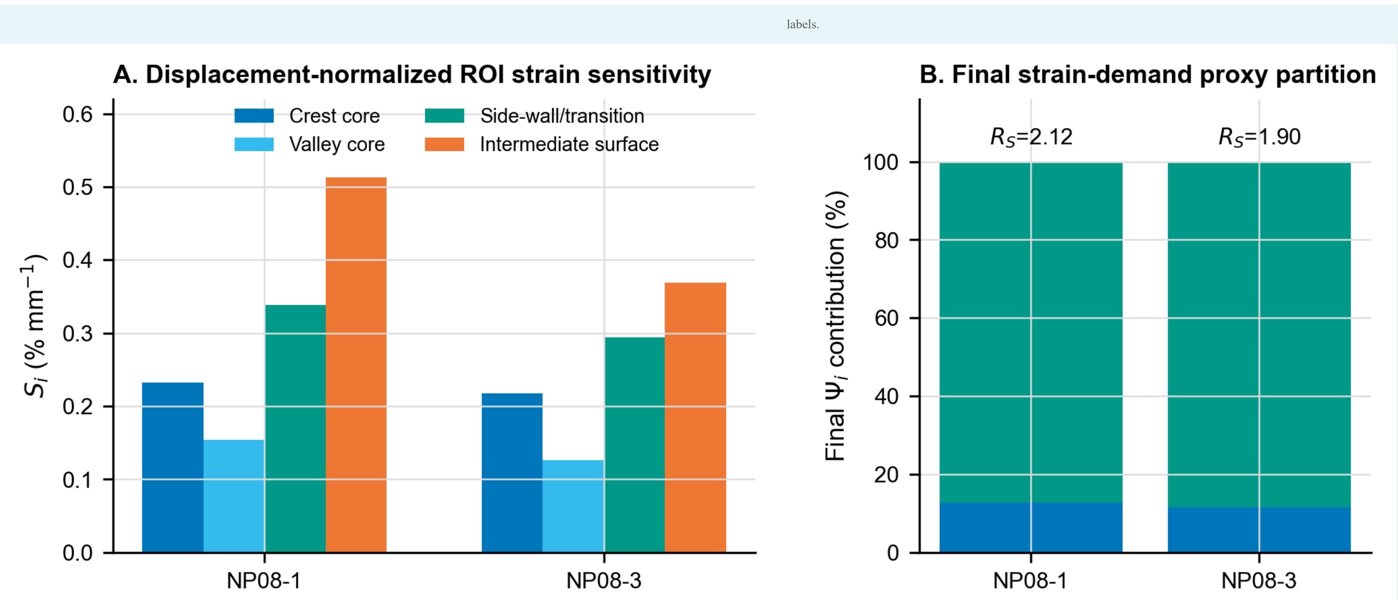
R1. Results: DIC/PLC Mechanism Evidence



DIC strain span follows imposed PLC mean displacement.



Initial geometry defines persistent ROI labels.

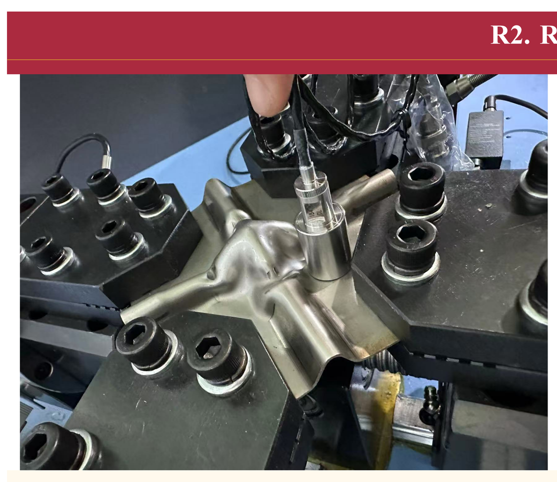


High-deformation ROIs carry most final strain demand.

Metric	NP08-1	NP08-3	Interpretation
Sensitivity ratio R_S	2.117	1.896	high-ROI response is about 2x core response
High-deformation area	54.28%	53.96%	deformation path occupies about half the map
Final strain-demand share	87.2%	88.4%	most demand stays in compliant regions

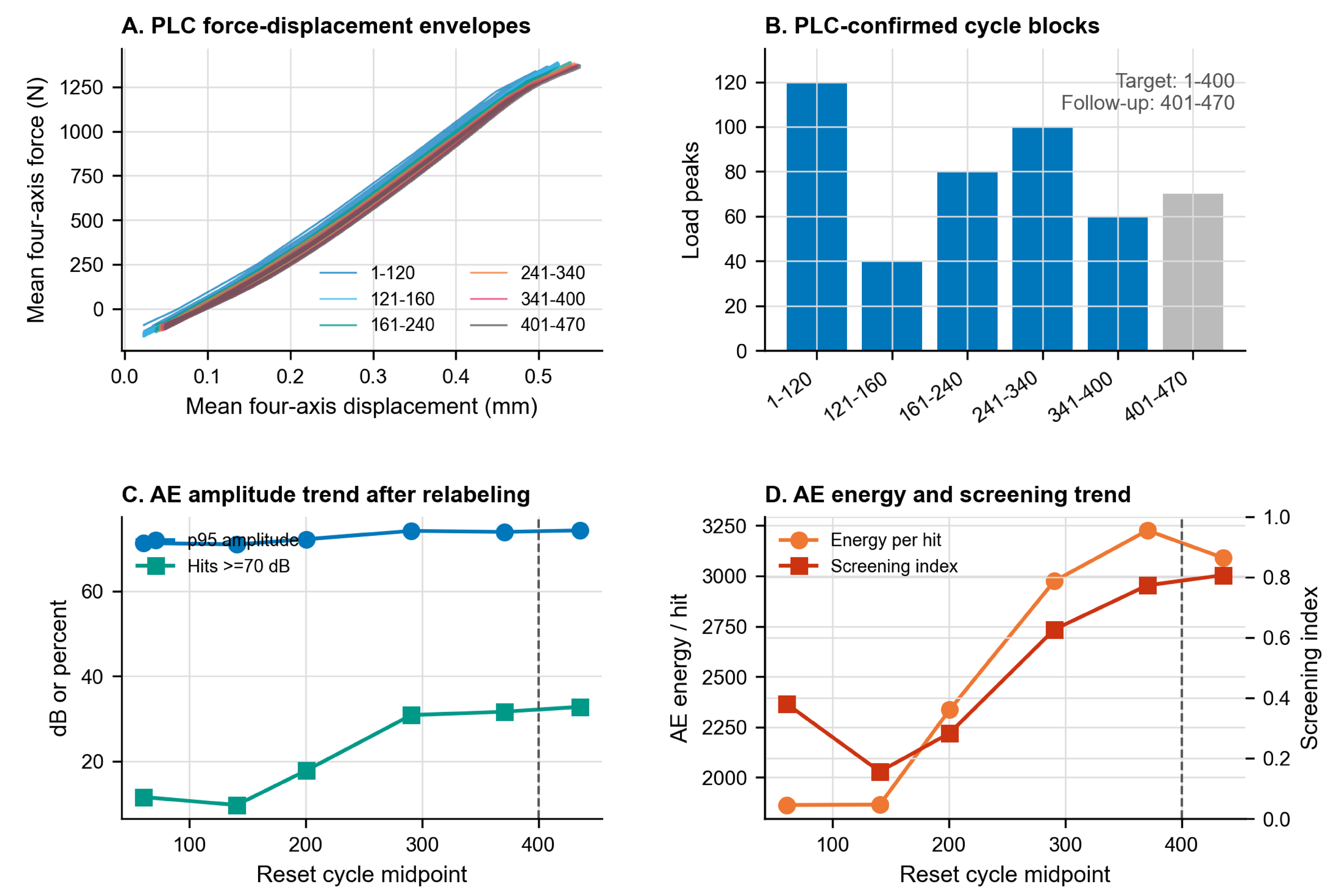
Mechanism result. Corrugated specimens show repeated ROI hierarchy: side-wall/transition and intermediate surfaces absorb the imposed displacement while crest/valley cores remain lower-demand regions.

R2. Results: AE Activity and Assembly Integrity Checks

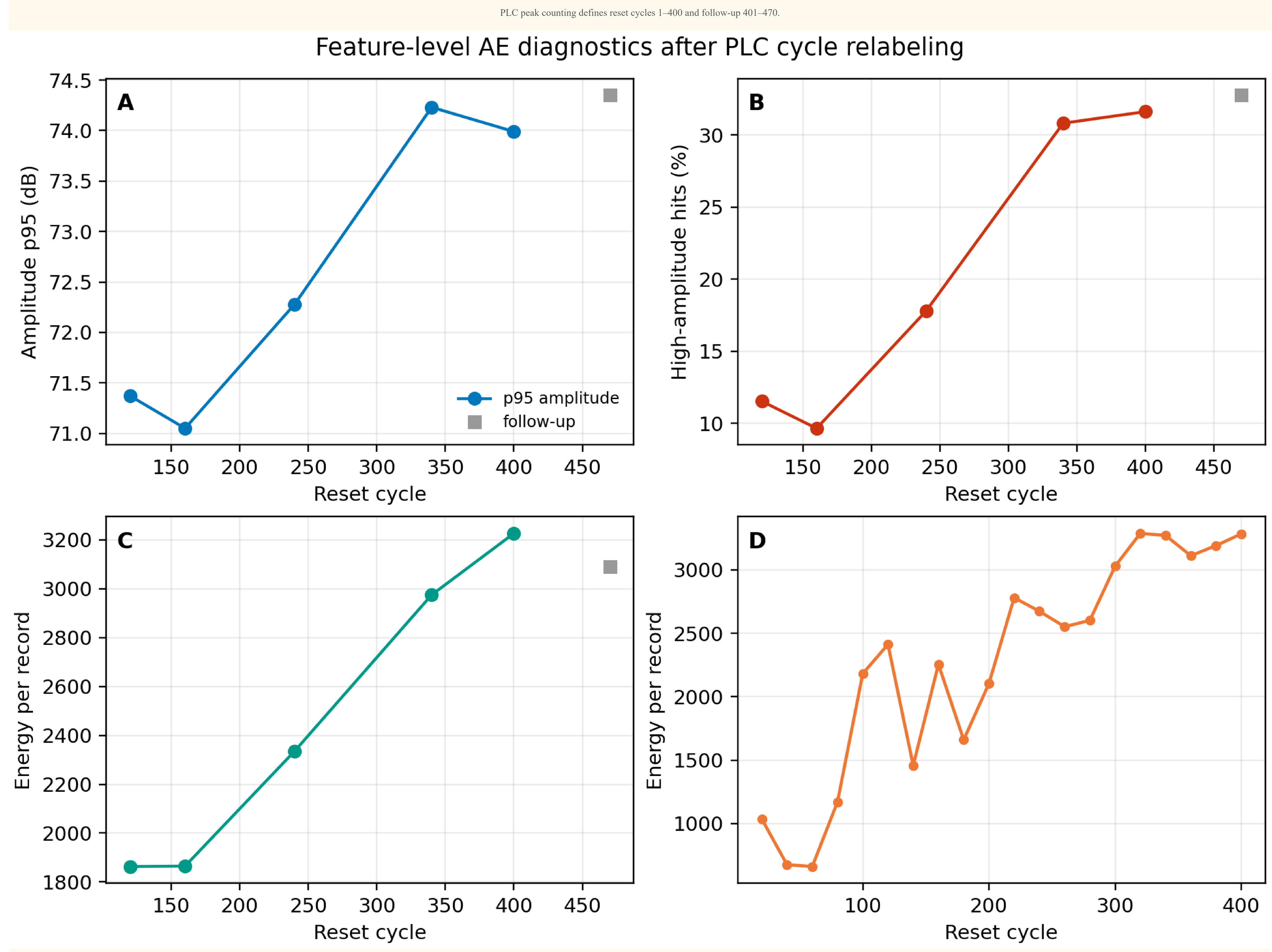


AE sensors were mounted during the same four-axis loading context.

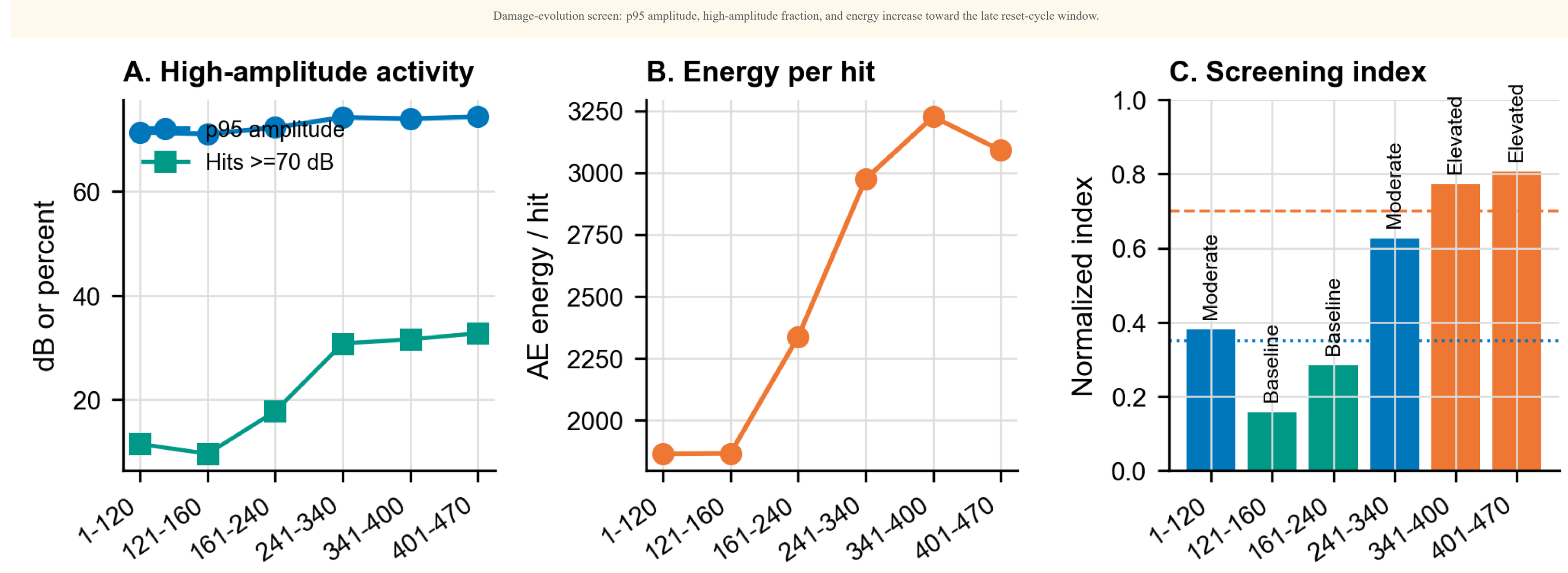
Rule	Interpretive use
Relabel cycles	PLC peak counting resets the AE coordinate before interpretation.
Track features	amplitude, high-amplitude fraction, and energy indicate cyclic activity.
Keep boundary	feature trends screen damage evolution; they do not alone prove cracks.



PLC peak counting defines reset cycles 1–400 and follow-up 401–470.



Damage-evolution screen: p95 amplitude, high-amplitude fraction, and energy increase toward the late reset-cycle window.

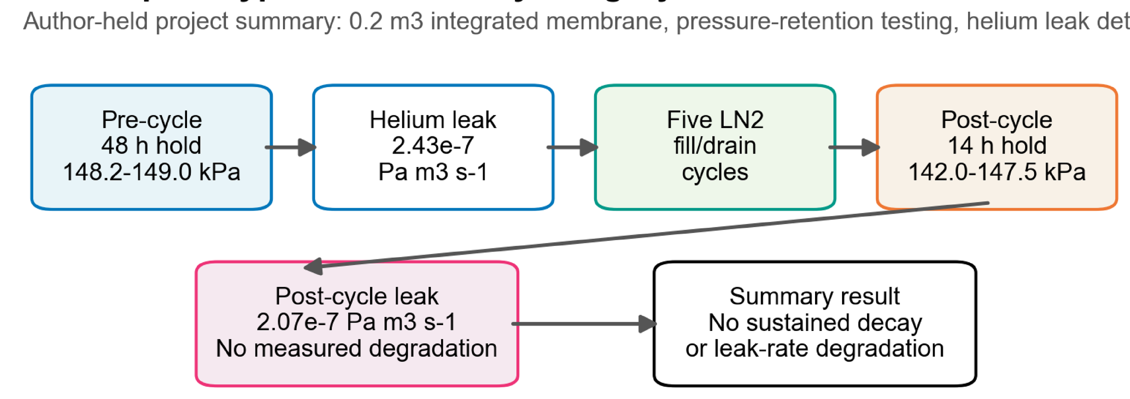


Screening index ranks cycles 341–400 as the inspection-priority window.

31.63%	3.23×10^3	**125 kHz**	**1.17M**
records above 70 dB	energy per hit	median frequency	feature records

Scaled-prototype LN2 assembly-integrity check

Author-held project summary: 0.2 m3 integrated membrane, pressure-retention testing, helium leak detection, and five LN2 filling/draining cycles.



Claim boundary: pressure-retention is the test method; the supported objective is preliminary assembly integrity, not LH2 qualification.

Pressure-hold and helium leak-rate checks are the test method; the objective is to verify assembled-membrane integrity before and after five LN_2 filling/draining cycles.

Check	Result
Pre-cycle hold	48 h at 148.2–149.0 kPa abs.
Post-cycle hold	14 h at 142.0–147.5 kPa abs.
Pre-cycle leak	2.43×10^{-7} Pa m^3 s^{-1}
Post-cycle leak	2.07×10^{-7} Pa m^3 s^{-1}

D. Discussion: What the Evidence Supports

Layer	Supports	Does not claim	Evidence panel
DIC/PLC	corrugation-driven strain redistribution	service lifetime	R1
AE	late-cycle inspection-priority window	crack growth proof	R2-AE
LN_2	preliminary assembly integrity	LH_2 qualification	R2-LN_2

- The strongest evidence is mechanism evidence from synchronized DIC/PLC and ROI mapping.
- AE is useful after relabeling because it ranks when inspection attention should focus.
- LN_2 pressure-retention testing is necessary but not sufficient for final LH_2 service readiness.

C. Conclusions and Next Gates

Conclusions

1. Corrugation redistributes strain into traceable geometry-defined ROIs.
2. The late AE reset window gives an inspection-priority signal.
3. Pre/post LN_2 pressure and leak checks support preliminary assembly integrity.

Next gates

1. Repeat coupon tests with ROI uncertainty.
2. Add cryogenic DIC and waveform-level AE.
3. Report full acceptance pressure/leak records under LH_2-relevant criteria.

ICEC/ICMC record	P1-061, Poster Session I, Exhibition Hall (111+112), June 23, 2026, 14:00–16:00
Claim boundary	The evidence supports staged validation under the tested conditions; completed LH_2 service qualification remains outside the present scope.

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